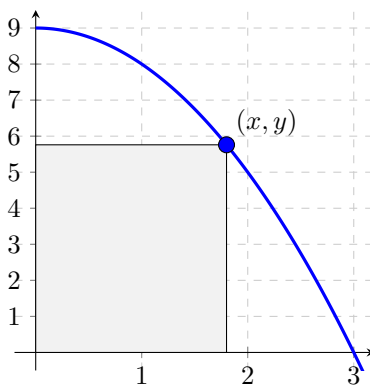


Problem 4

Problem 1 A rectangle in the first quadrant is constructed by taking a point (x, y) on the graph of the function $f(x) = 9 - x^2$, drawing a line segment vertically downward to the x -axis and a line segment horizontally leftward to the y -axis, as in the picture below. Denote the **length of the base** (along the x -axis) of the rectangle by x , the **height of the rectangle** (along the y -axis) by y (in meters), and the **PERIMETER** of the rectangle by P . When $x = 2$ m (and only at that moment), the height y is shrinking at a rate of $\frac{1}{5}$ m/s. Find the value of $\left[\frac{dP}{dt}\right]_{x=2m}$ by performing the steps below.



- Find a formula** for the height, y , of the rectangle as a function of x . (This is denoted as $y(x)$.)
- Find the value** of $\left[\frac{dx}{dt}\right]_{x=2m}$.
- Find a formula** for the perimeter, P , of the rectangle as a function of x . (This is denoted as $P(x)$.)
- Find the value** of $\left[\frac{dP}{dt}\right]_{x=2m}$.
- Write a sentence to explain** what the value found in (d) means about the rectangle. (Don't forget UNITS.)